LAB 1 TỔNG HỢP
Art Gallery Manager App

1. Project Overview

The **Art Gallery Manager Application** is a mobile application that helps users manage artworks stored in a digital gallery.

The application allows users to:

- register a new account
- login to the system
- manage session state
- view artworks
- add new artworks
- edit artworks
- delete artworks
- search artworks
- filter artworks by category or year
- mark artworks as favorite
- view favorite artworks
- view gallery statistics

All application data must be stored **locally** using **SharedPreferences**, **SQLite** or **Hive**.

The application must provide a **clean, modern, and user-friendly interface**.

2. Learning Objectives

After completing this assignment, students should be able to:

- build multi-screen mobile apps with Flutter
- implement user authentication flow
- manage user sessions
- perform CRUD operations with local database
- implement search and filter functionality
- implement relationships between data entities
- calculate and display statistical summaries
- design user-friendly mobile UI

3. Technology Requirements

Component	Requirement
Framework	Flutter
Language	Dart
Database	SQLite or Hive
IDE	Android Studio / VS Code
UI Framework	Material Design

4. Project Naming

Students must use the following project name:

`galleryManager_StudentCode`

Example:

`galleryManager_SE123456`

5. Data Models

5.1 User Model

Field	Type	Description
id	int	User ID
username	String	Login username
email	String	User email
password	String	User password
createdAt	String	Account creation date

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5.2 Artwork Model

Field	Type	Description
id	int	Artwork ID
title	String	Artwork title
artist	String	Artist name
year	String	Creation year
category	String	Artwork category
description	String	Artwork description
createdBy	int	User ID (owner)

5.3 Favorite Model

Field	Type	Description
id	int	Favorite ID
userId	int	User ID
artworkId	int	Artwork ID
createdAt	String	Favorite date

6. Database Relationships

```
User
  |
  | 1
  |
  | n
Artwork
  |
  | 1
  |
  | n
FavoriteArtwork
```

Meaning:
• One user can create multiple artworks
• One user can favorite multiple artworks

7. Application Screens

The application must include the following screens:

1. Register Screen
2. Login Screen
3. Home Screen
4. Add Artwork Screen
5. Artwork Detail Screen
6. Edit Artwork Screen
7. Favorite Artworks Screen

8. Functional Requirements

Task 1 – User Registration

Create a **Register Screen**.
Fields:

```
Username
Email
Password
Confirm Password
```

Requirements:
• Validate that all fields are filled
• Password and Confirm Password must match
• Username must be unique
After successful registration:

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```
        Save user data to database
      Navigate to Login Screen
```

Task 2 – User Login

Create a **Login Screen**.
Fields:

```
        Username
      Password
```

Requirements:
• Validate input fields
• Verify credentials from database
If login succeeds:

```
      Create user session
      Navigate to Home Screen
```

If login fails:

```
        Show error message
      Invalid username or password
```

Task 3 – Session Management

The application must remember login status.
When the app starts:

```
      Check session storage
```

If session exists:

```
        Open Home Screen
```

If session does not exist:

```
        Open Login Screen
```

Session may be stored using:
• SharedPreferences
• Hive

Task 4 – Logout

Add a **Logout button** on Home Screen.
When clicked:

```
      Clear session data
      Return to Login Screen
```

Display notification:

```
        Logged out successfully
```

Task 5 – Home Screen (Artwork List)

The Home Screen must display a list of artworks.
Display information:

```
      Title
      Artist
      Year
```

Requirements:
• Load data from database
• Display using ListView.builder
• Show gallery header image/logo
Navigation options:

```
      Add Artwork
      Artwork Detail
      Logout
      Favorite Artworks
```

Task 6 – Add Artwork

Create **Add Artwork Screen**.
Fields:

```
      Title
      Artist
```


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```
Year
Category
Description
```

Requirements:
• Artwork ID must auto-increment
• Save artwork into database
After saving:

```
Return to Home Screen
Update artwork list
```

Task 7 – Artwork Detail

When user taps an artwork:
Open **Artwork Detail Screen**.
Display:

```
Title
Artist
Year
Category
Description
```

Options:

```
Edit Artwork
Delete Artwork
Add to Favorite
```

Task 8 – Update Artwork

Users can edit artwork information.
Requirements:
• Modify any field
• Save changes to database
• Updated data must appear immediately on Home Screen.

Task 9 – Delete Artwork

Add **Delete button** on Artwork Detail Screen.
Before deletion:
Show confirmation dialog.
Example:

```
Are you sure you want to delete this artwork?
```

After deletion:

```
Remove artwork from database
Update artwork list
```

Show notification using SnackBar.

Task 10 – Search Artworks

Add a **Search Bar** on Home Screen.
Requirements:
• Search artworks by **Title**
• Filter results dynamically while typing.
Example:

```
Search: Sun
```

Results:

```
Sunrise Over the Sea
Sunlight Landscape
```

Task 11 – Filter Artworks

Allow filtering by:

```
Year
Current year
Last 5 years
All years
```


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Category

```
Abstract
Realism
Landscape
Portrait
```

Suggested UI components:

```
DropdownButton
Filter Chips
RadioButtons
```

Task 12 – Favorite Artworks

Users can mark artworks as **Favorite**.
Add button:

```
Add to Favorite
```

Requirements:
• Save favorite record in database
• Prevent duplicate favorites.

Task 13 – Favorite Artworks Screen

Create **Favorite Artworks Screen**.
Display list:

```
Title
Artist
Year
```

Users can:

```
Remove artwork from favorites
```

Task 14 – Gallery Statistics

Display gallery statistics on Home Screen.
Example:

```
Gallery Statistics
------------------
Total artworks: 25
Favorite artworks: 8
Categories: 4
```

Statistics must update automatically when:

```
Artwork added
Artwork deleted
Favorite added
Favorite removed
```


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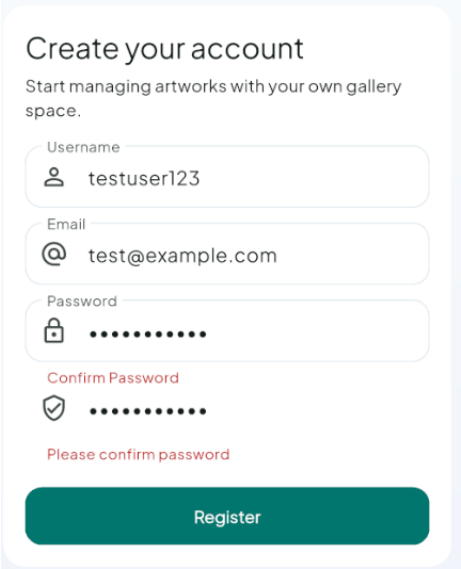

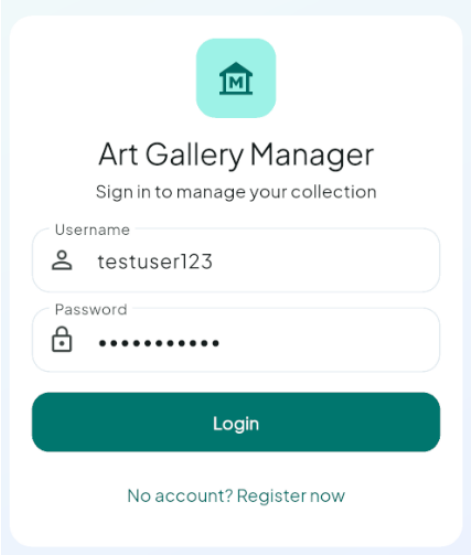

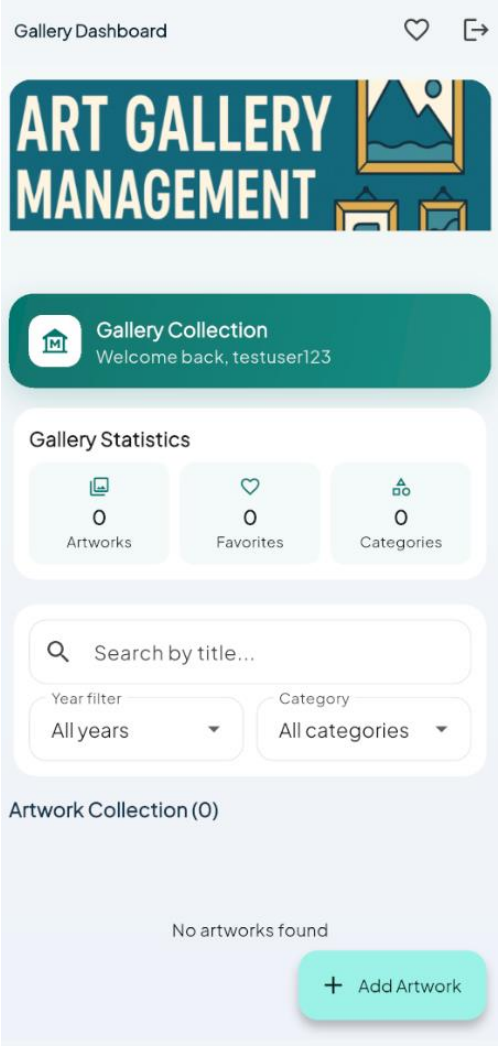

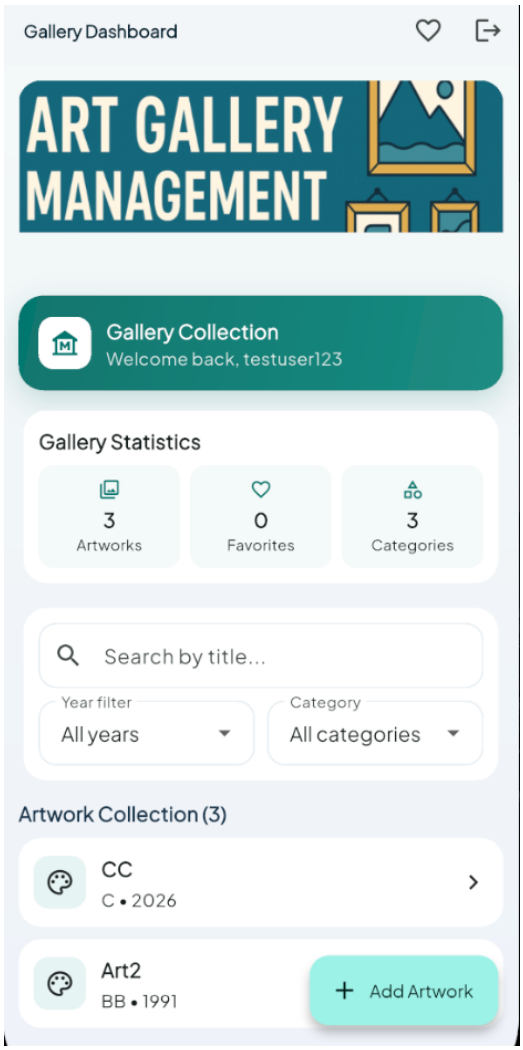

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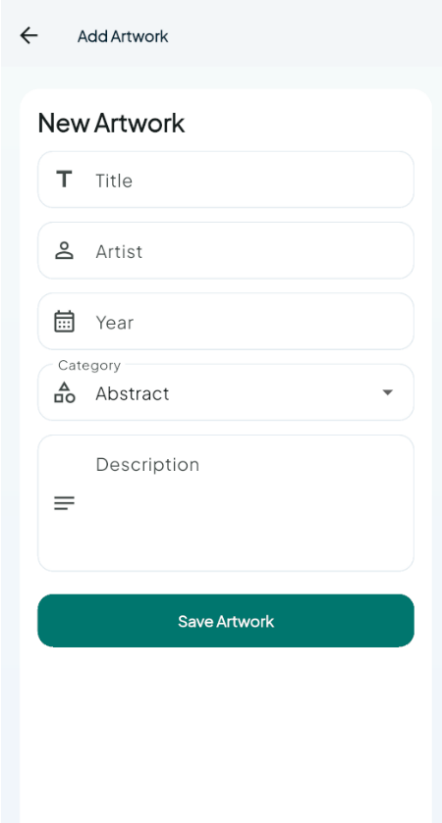

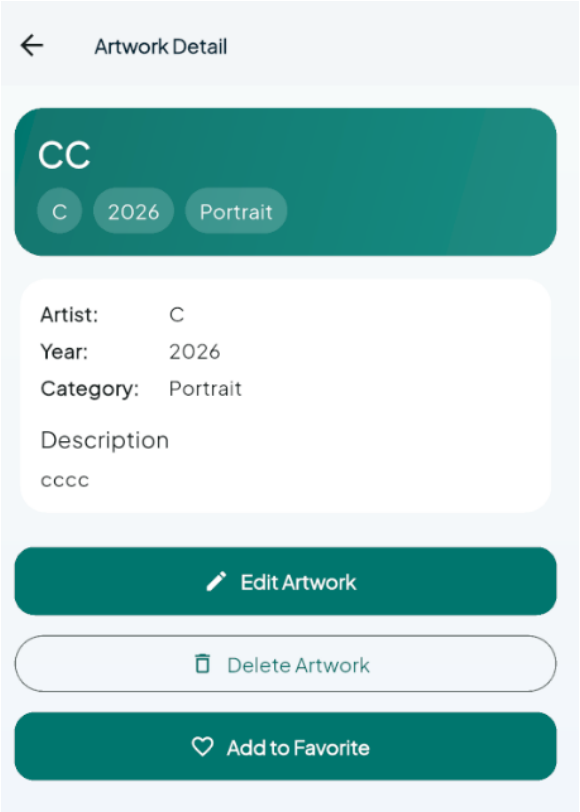

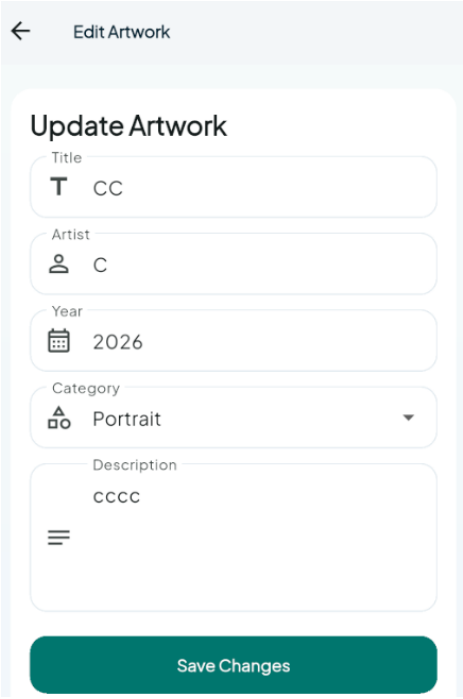

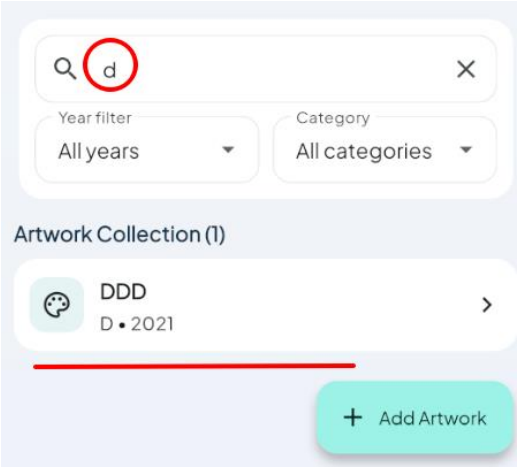

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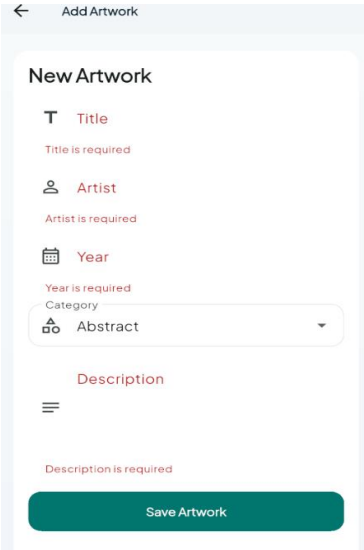

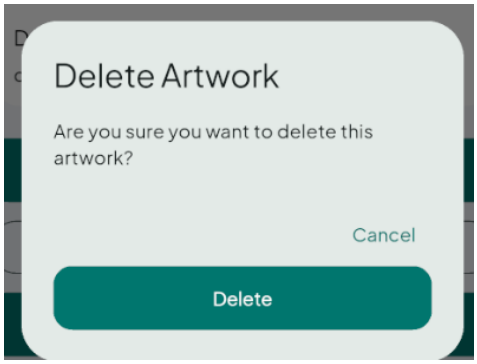

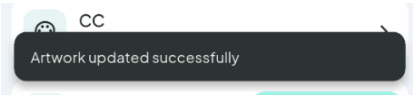

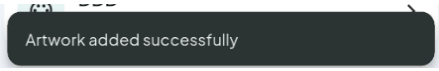

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